Why graph

ERC starts from a connectivity graph , instances, nets, and pin edges

The idea

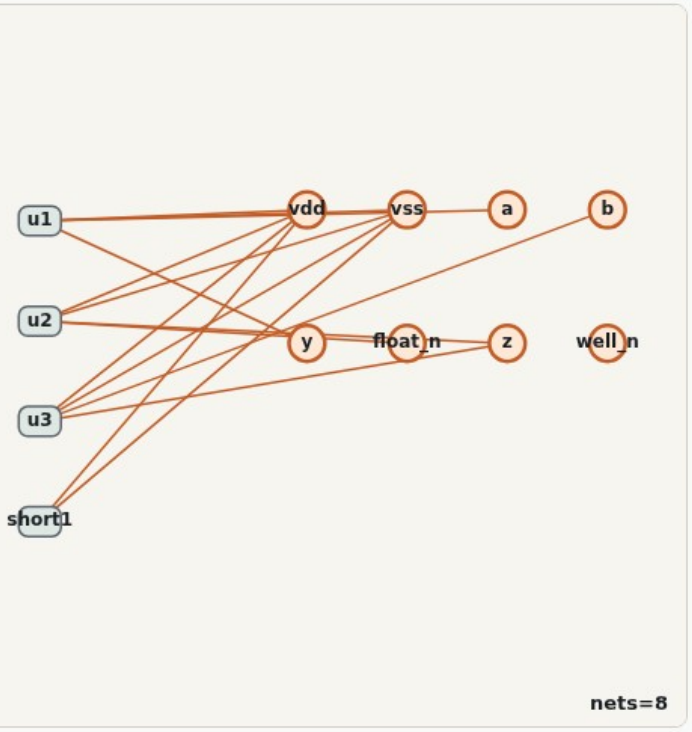
- GOLDEN: 8 nets, 4 instances, 14 pin edges

Nets

NETLIST CONNECTIVITY GRAPH

Nets

Step 1 / 5 · nets · module01-01-netlist-graph



Explanation

Eight nets: supplies, ports, float_n, z, well_n.

- nets

n=8

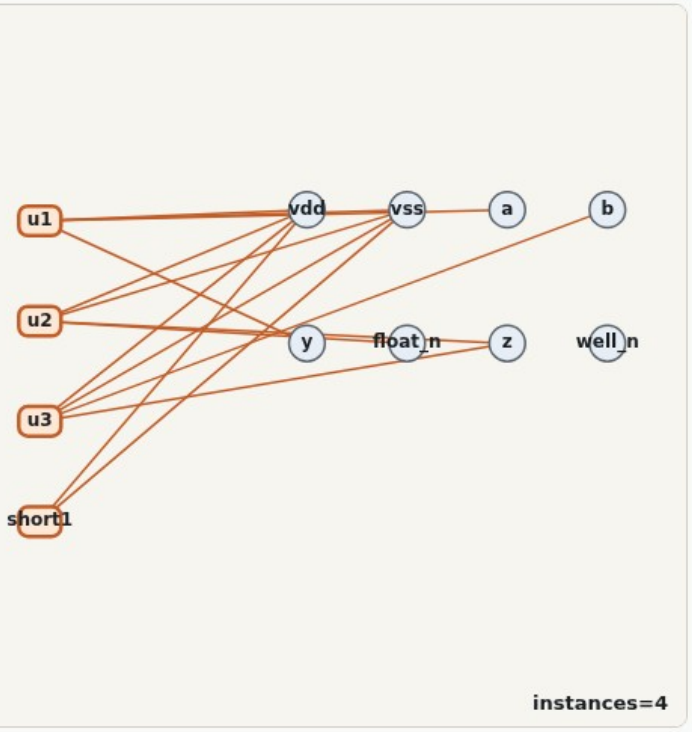
Nets

Instances

NETLIST CONNECTIVITY GRAPH

Instances

Step 2 / 5 · inst · module01-01-netlist-graph



Explanation

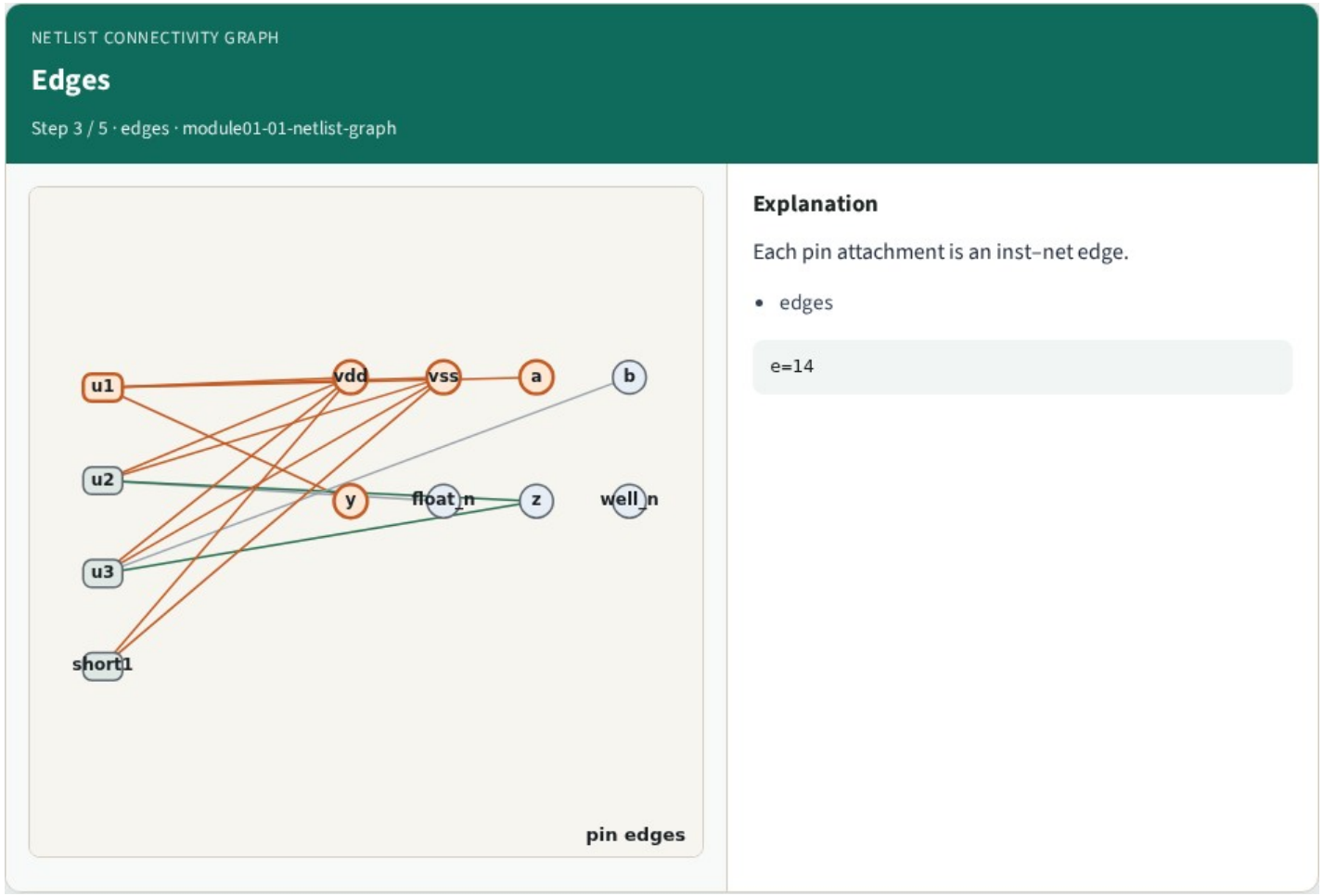
Four instances: u1, u2, u3, short1.

- instances

```
i=4
```

Instances

Edges


NETLIST CONNECTIVITY GRAPH
Edges
Step 3 / 5 · edges · module01-01-netlist-graph
u1
u2
u3
short1
vdd
vss
a
b
y
float_n
z
well_n
pin edges
Explanation
Each pin attachment is an inst–net edge.
edges
e=14

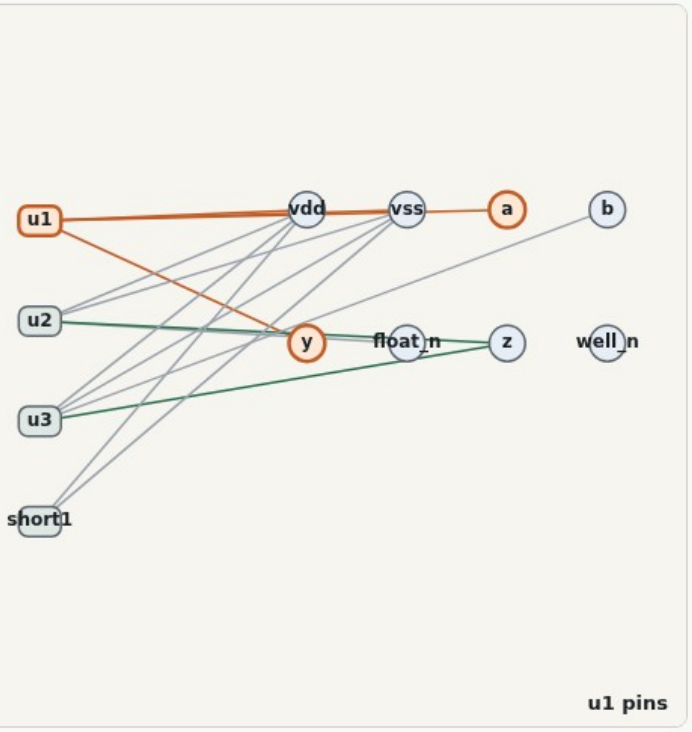
Edges

Walk

NETLIST CONNECTIVITY GRAPH

Walk

Step 4 / 5 · walk · module01-01-netlist-graph



Explanation

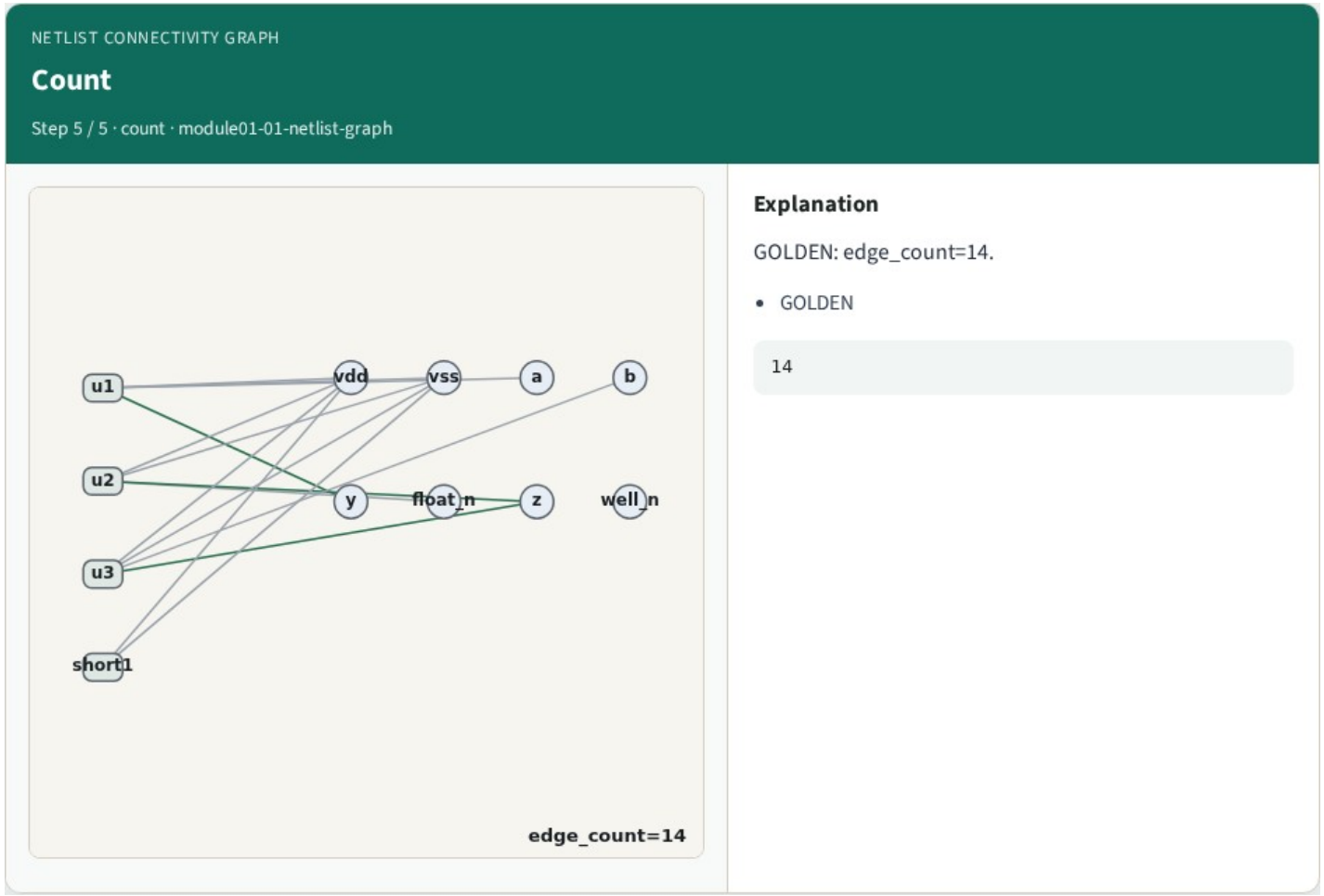
Walk pins on u1 INV: A,Y,VDD,VSS.

- u1

```
A/Y/VDD/VSS
```

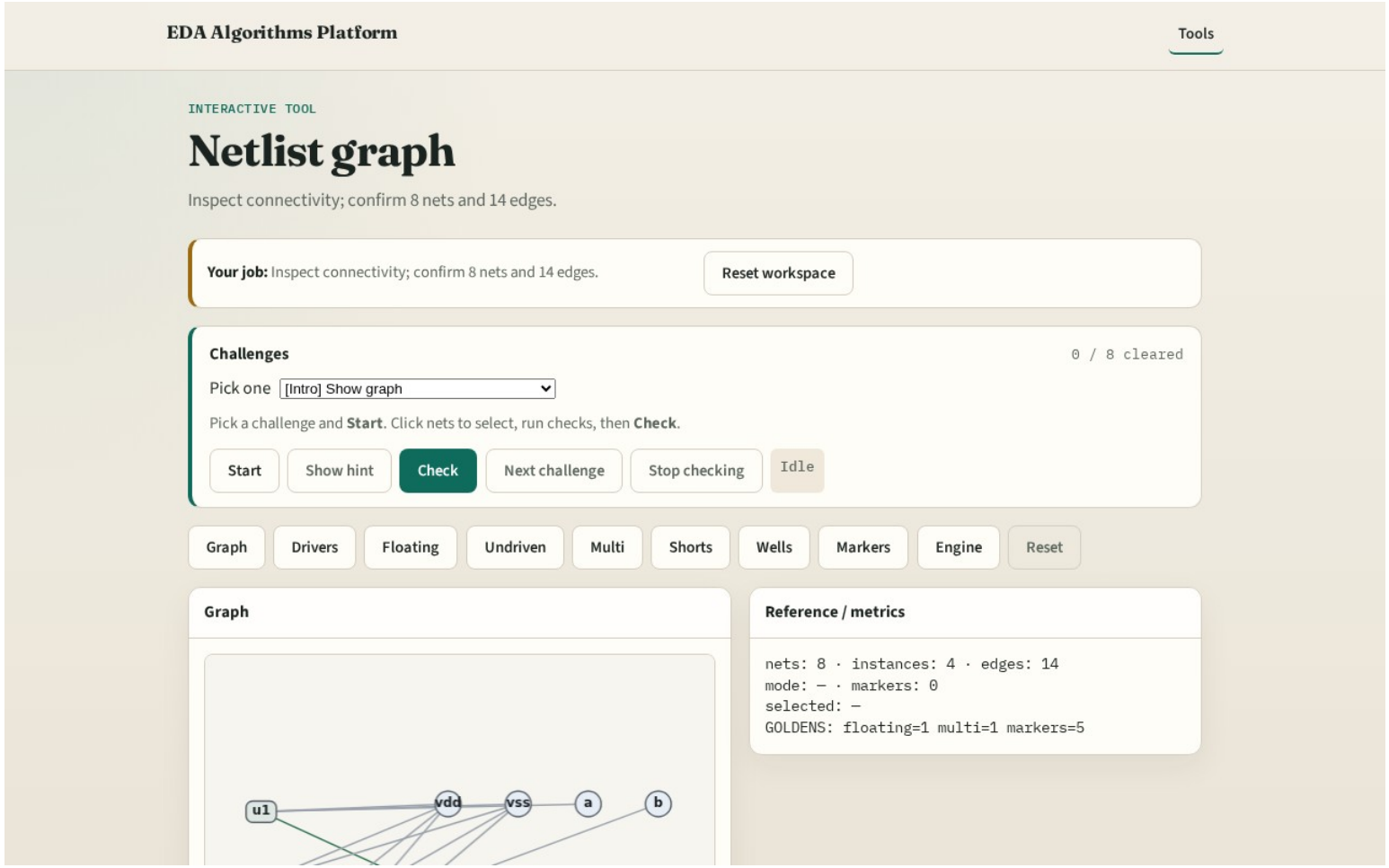
Walk

Count


NETLIST CONNECTIVITY GRAPH
Count
Step 5 / 5 · count · module01-01-netlist-graph
u1
u2
u3
short1
vdd
vss
a
b
y
float_n
z
well_n
edge_count=14
Explanation
GOLDEN: edge_count=14.
GOLDEN
14

Count

Browser lab track



Browser lab starter

Implement track

- Call design_stats() and build_connectivity(); assert net_count==8, edge_count==14

Pitfalls

- Counting wells as instances
- Double-counting supply pins

Your turn

- Next: classify drivers vs loads